

Automated Disease Identification in Tea and Betel Leaf Crops using Improvised MobileNet V3

Presented by-

BISHAL SARMAH (23MSD7048)

VIVEK SHARMA (23MSD7013)

M Sc. Data Science

Dept. Of Mathematics, School Of Advanced Sciences, VIT-AP University,
Andhra Pradesh, 522241, India

ABSTRACT

In this work, we propose a lightweight and efficient deep learning model for leaf disease detection using MobileNetV3 enhanced with Efficient Layer Attention (ELA). The primary objective is to develop a model that offers high accuracy while being suitable for deployment on resource-constrained devices such as smartphones or edge systems. Through extensive experiments, we observed that our proposed model outperforms several other lightweight architectures, including baseline MobileNetV3, MobileNetV3 with partial fine-tuning, MnasNet, and a custom CNN model. Among all tested models, MobileNetV3 + ELA achieved the highest classification accuracy on the tea and betel leaf disease dataset. This model not only ensures accurate detection but also maintains computational efficiency, making it ideal for use in real-time applications. The solution is particularly beneficial for small and marginal farmers, enabling them to identify plant diseases early and take timely actions without the need for expensive or complex systems.

INTRODUCTION

Agricultural & Economic Significance

India stands as the world's second-largest producer of tea, with Assam playing a crucial role by contributing over 50% of the country's tea exports. In Assam, small farmers engage in mixed cultivation, growing both tea and betel vine on the same plots of land. This practice makes both crops vital to the region's agricultural economy and essential for the livelihoods of local communities

Motivation

- Mixed Crop Cultivation in Assam: Small farmers in Assam commonly grow tea and betel vine crops together, but existing disease detection systems focus on single-crop scenarios, creating a gap for practical, localized solutions.
- Challenges for Small Farmers: Manual inspections by experts are costly, slow, and inaccessible for resource-constrained farmers, leading to delayed diagnoses, crop losses, and overuse of pesticides.
- Need for On-Device AI: Farmers rely on smartphones; a lightweight, accurate model for combined tea and betel leaf datasets is critical for real-time, on-device disease diagnosis.

OBJECTIVES

Research Objectives

1. Dataset & Model Optimization:

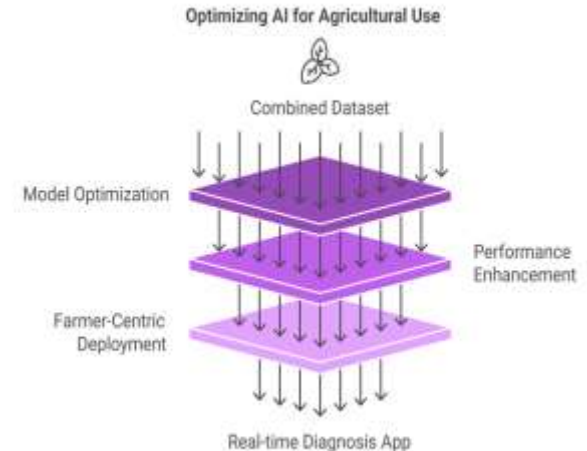
- Curate a combined dataset of tea and betel leaf diseases, reflecting Assam's mixed-crop reality.
- Experiment with small pre-trained models (MobileNet variants, MnasNet-A1, Custom CNN, EfficientNet-B0) to balance accuracy and parameter efficiency.

2. Performance Enhancement:

- Integrate attention mechanisms (Coordinate Attention [CA], Efficient Local Attention [ELA]) into the best-performing baseline model to improve disease localization.
- Ensure the final model achieves <5M parameters for smartphone compatibility while maintaining >90% accuracy.

3. Farmer-Centric Deployment:

- Design a user-friendly mobile interface for real-time diagnosis, prioritizing offline functionality and low data usage.



Methodology Overview

Data Collection

Two different datasets were collected:

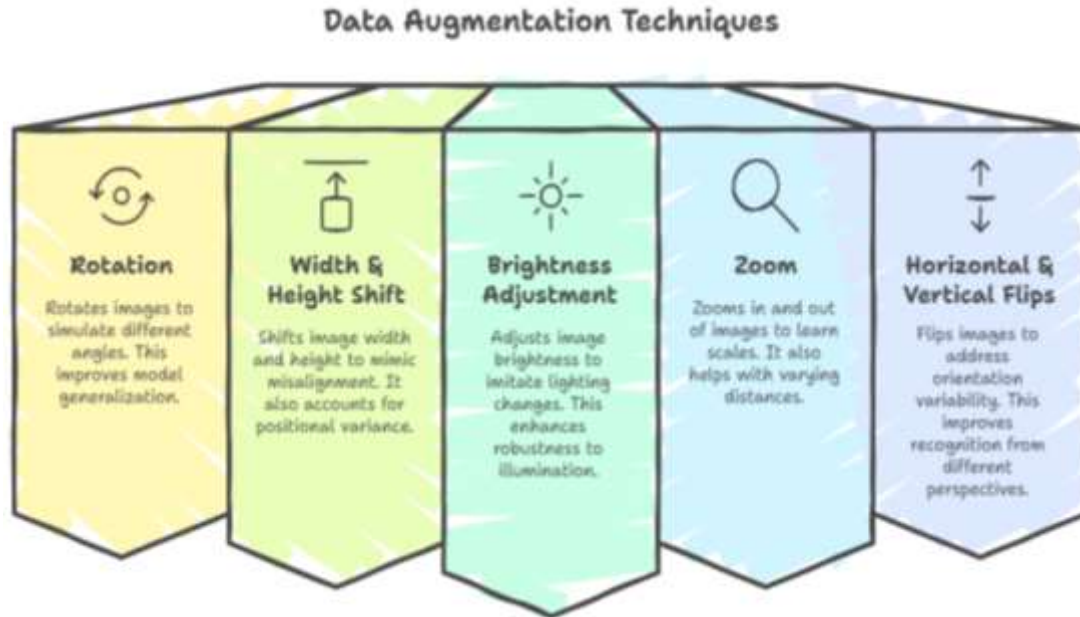
- Tea Leaves disease: This dataset is sourced from **Kaggle** which is a part of the study performed by Dutta et al. (2023). It contains **5877** images of six different diseases (e.g. - **red spot**, **helopelthis** etc.) found in tea leaves. The images belong to various Tea estates in the state of **Tripura** in **India**.
- Betel Leaves disease: This dataset is collected from the popular dataset site, **Mendeley Data**. It contains **1000** images of four different types of disease (e.g.- **bacterial leaf**, **dried leaf**, **fungal brown spot** including the **healthy leaf**) found in betel vine leaves. The image data belongs to the Tea estates of **Camperhat**, **Raipur** and **Lakshmipur** in **Bangladesh**.

The climate of the location of both the datasets is identical to that of **Assam** i.e. **Tropical Monsoon Climate**.

Methodology Overview

Data Preprocessing

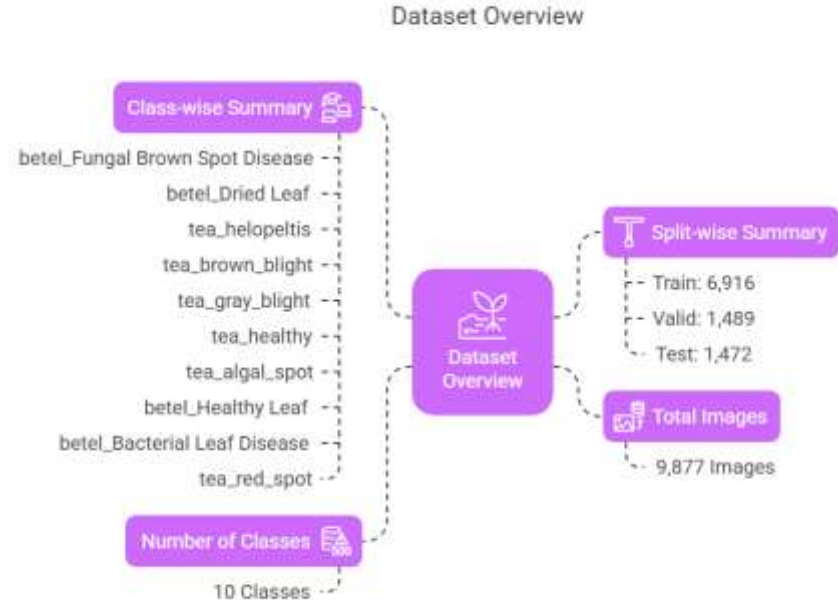
Augmentation: To enhance model robustness and prevent overfitting, several data augmentation techniques were used. The Betel leaf diseases, each containing 250 images were augmented to 1000 each.



Methodology Overview

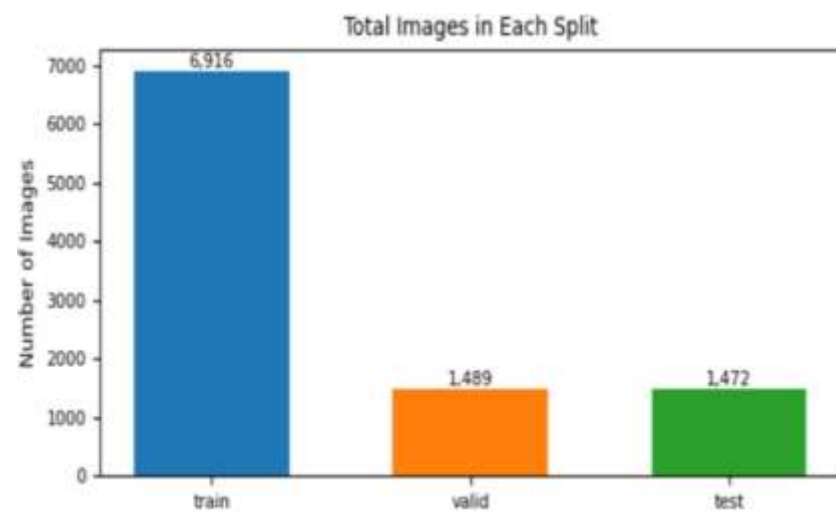
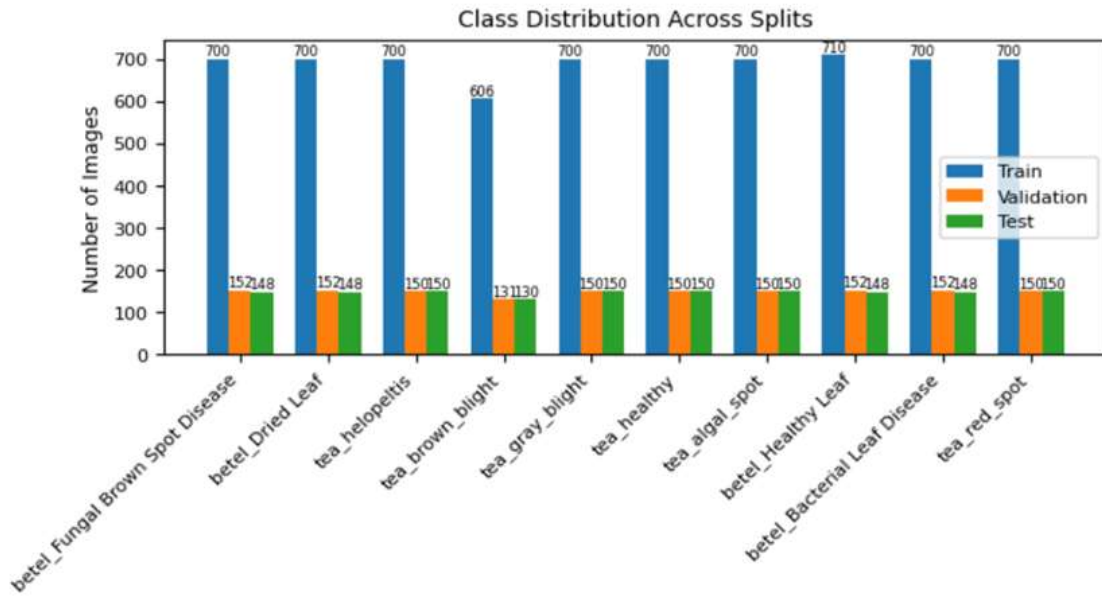
Data Preprocessing

- Data Splitting: The combined dataset (tea + betel) was divided into three subsets:
 - Training Set: 70%
 - Validation Set: 15%
 - Testing Set: 15%
- Resize: All images of both datasets were resized to a uniform dimension of $256 \times 256 \times 3$ pixels to ensure compatibility with the neural network.
- Pixel Scaling: Pixel values were normalized to the range $[0,1]$ for faster convergence during training.
- Class Encoding: Disease classes were converted into numerical format.



Methodology Overview

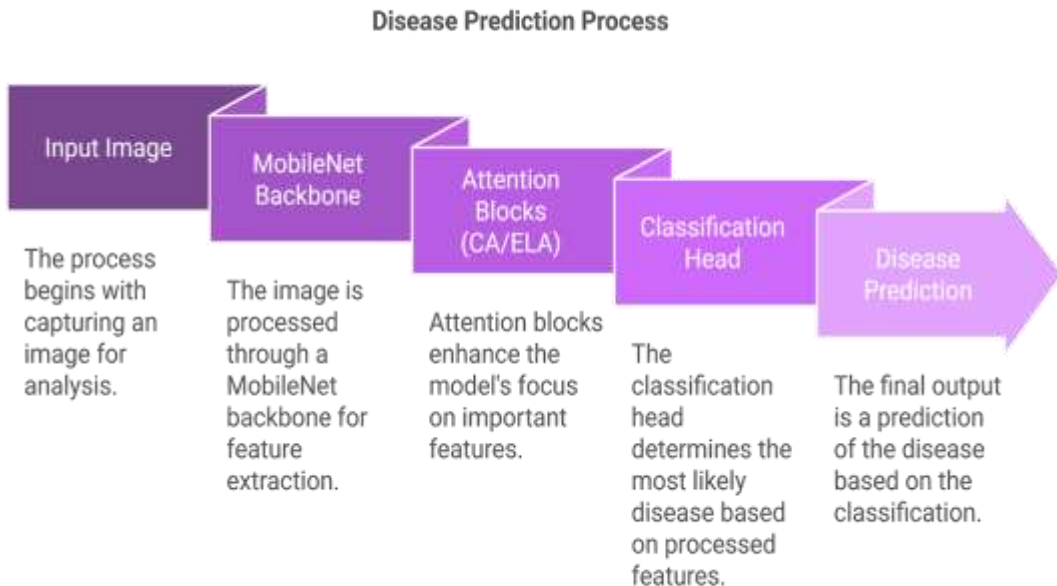
Dataset Summary



Model Selection

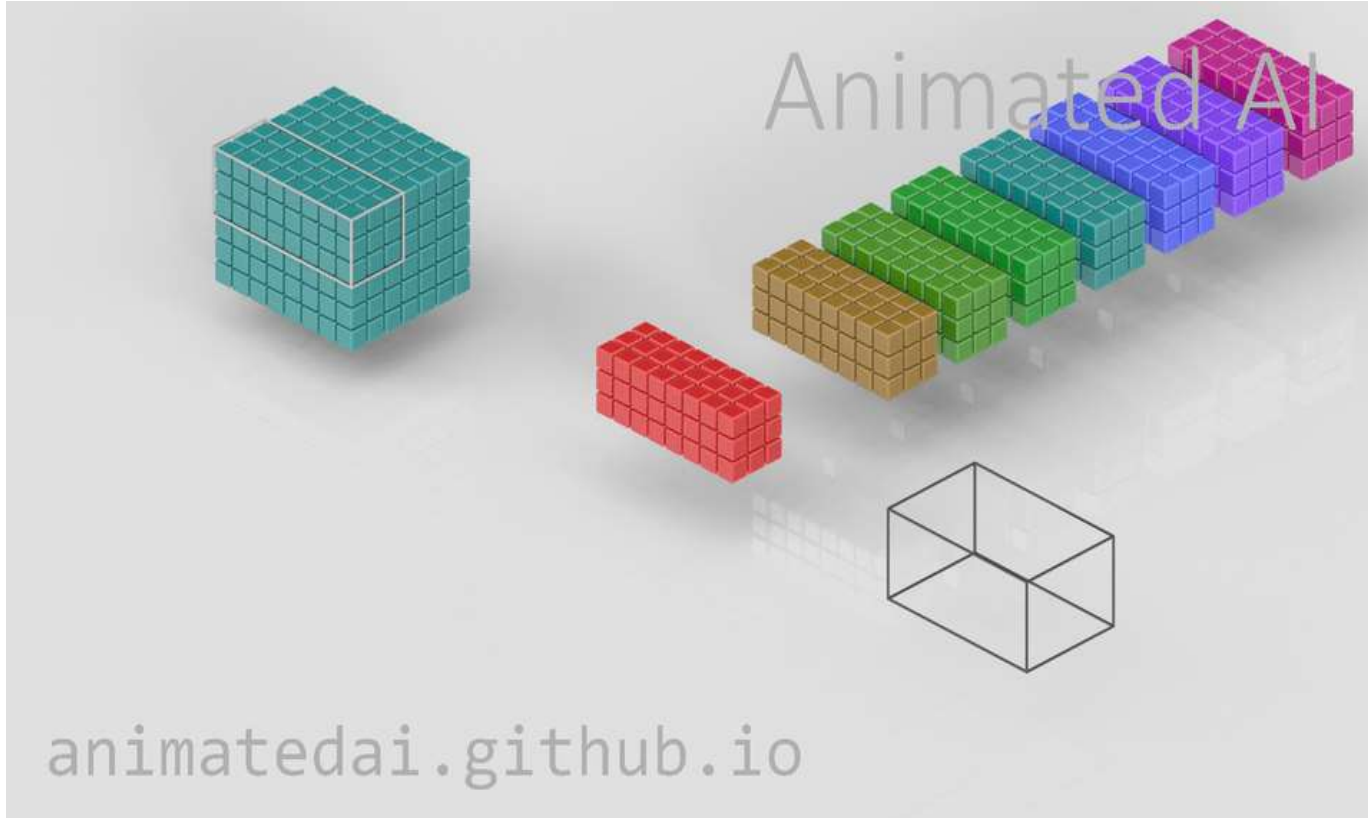
Why MobileNet?

- Depthwise & Pointwise Separable Convolutions (efficiency).
- Optimized for mobile (low parameters, fast inference).
- Proven on ImageNet; adaptable to small datasets.
- Excel in agricultural imaging-achieving up to 99.53 % accuracy on PlantVillage disease classification-while remaining compact enough for fast on-device inference



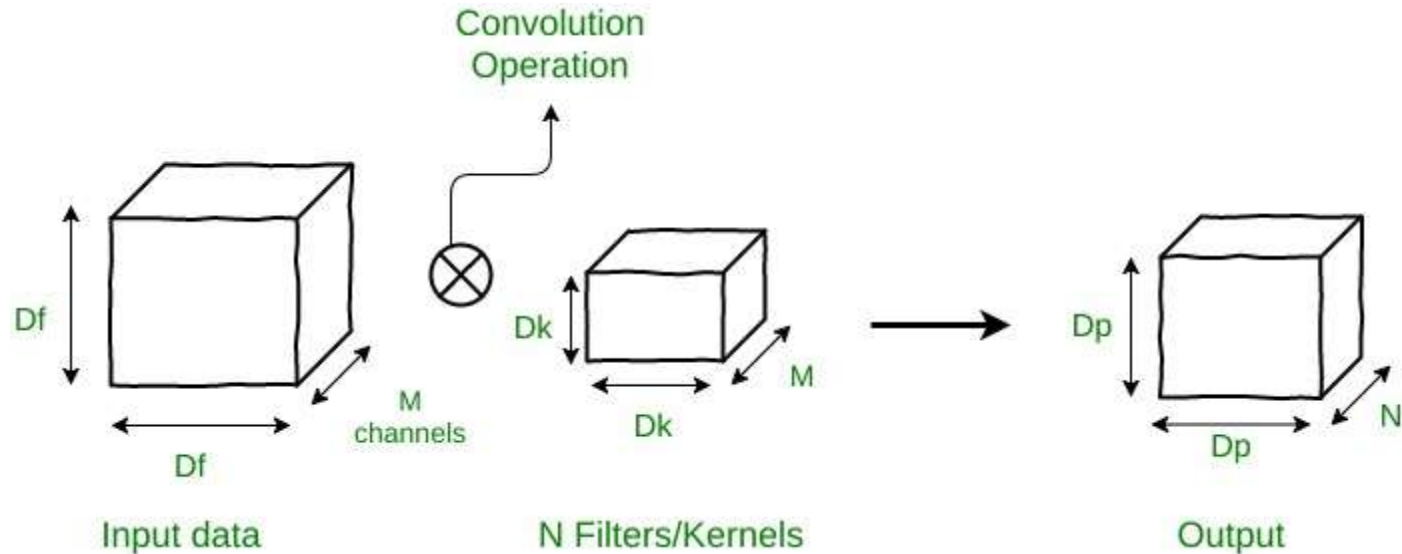
How MobileNet Reduces Computational Cost

Standard Convolution Operation



Animation from-
animatedai.github.io

Computation Cost

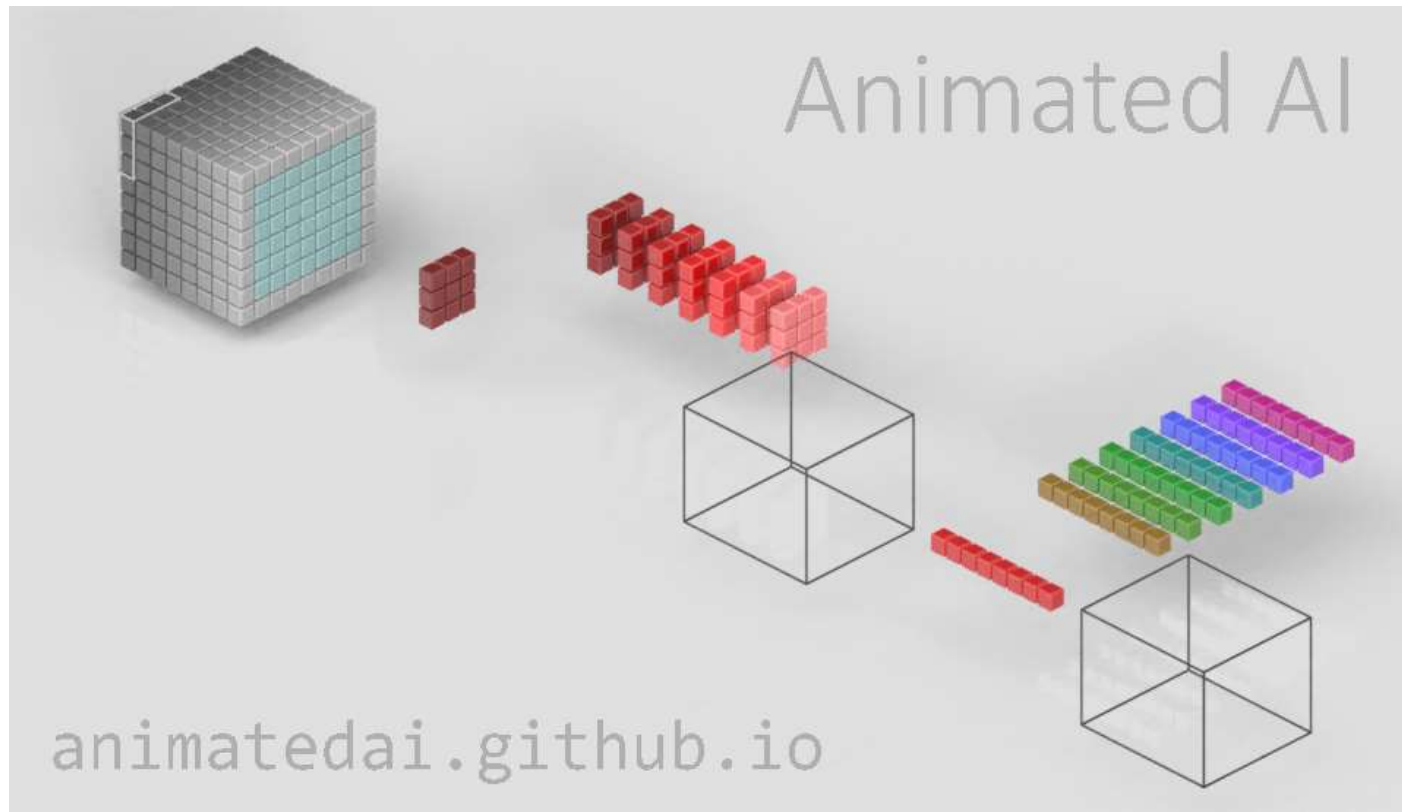


Computational Cost (Multiplications):

- Cost per filter pass: $D_k \times D_k \times M$
- Total Cost: $N \cdot D_p^2 \cdot D_k^2 \cdot M$

How MobileNet Reduces Computational Cost

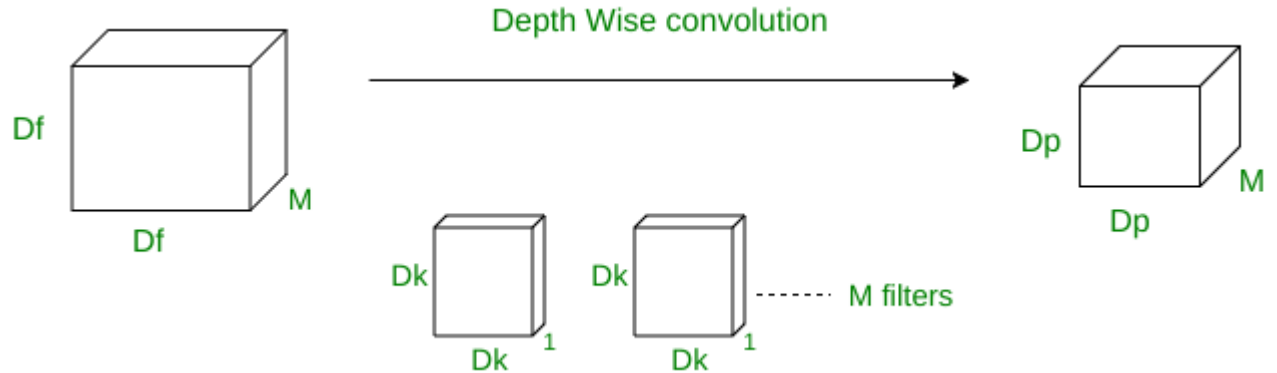
Depthwise Separable Convolution



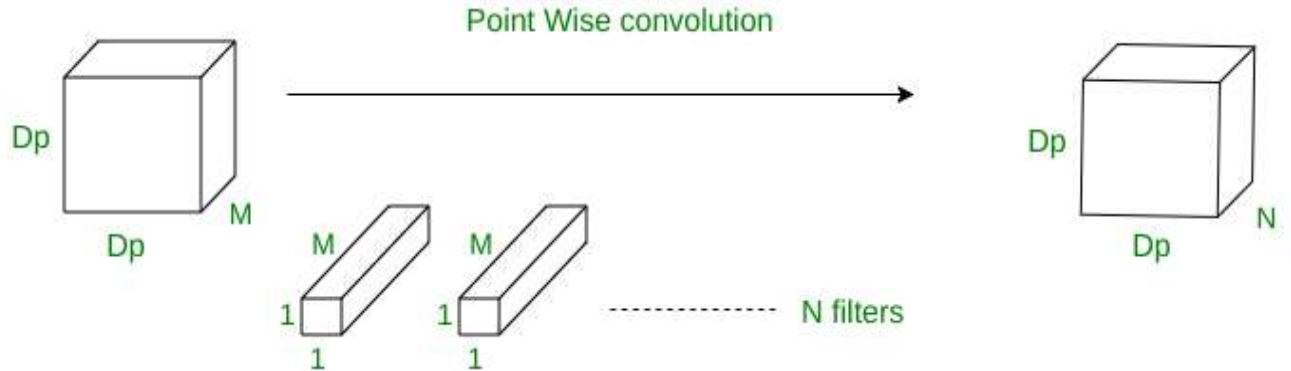
Animation from-
animatedai.github.io

Computation Cost

$$\text{Cost: } M \cdot D_k^2 \cdot D_p^2$$



$$\text{Cost: } M \cdot D_p^2 \cdot N$$



Computation Cost

Total Cost & Comparison

- Total Cost (Depthwise Separable): $M \cdot D_p^2 \cdot (D_k^2 + N)$
- Efficiency Gain:

Ratio (R) =

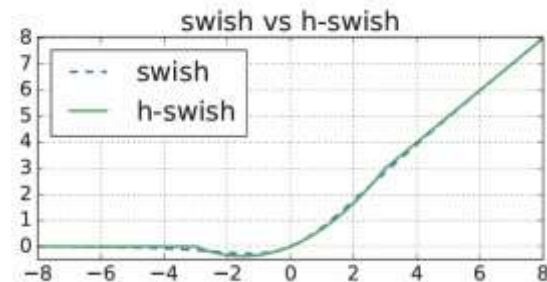
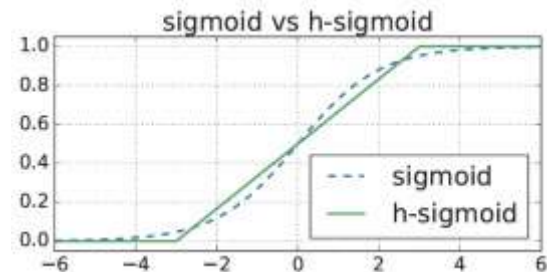
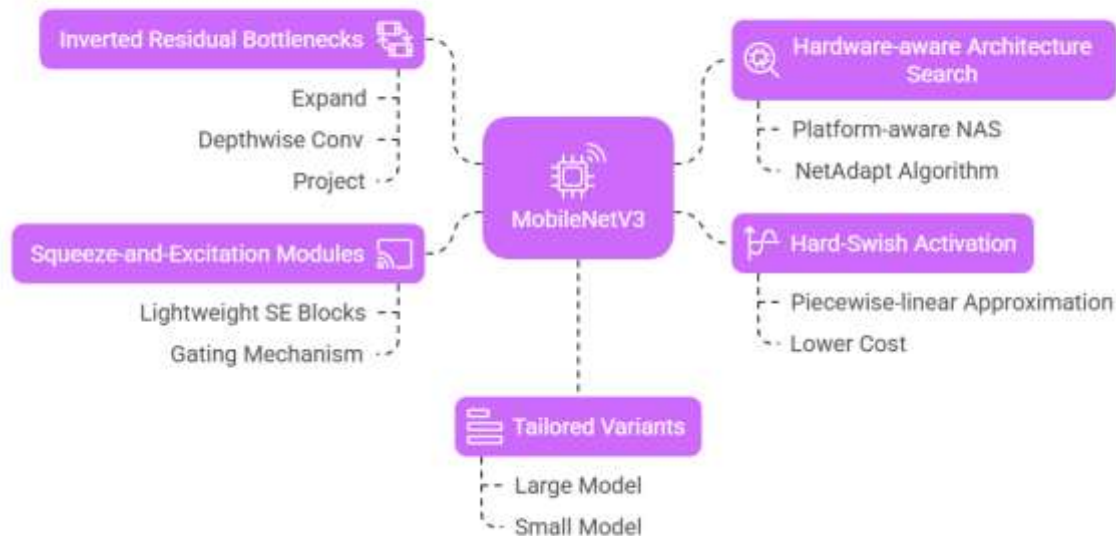
Cost of Standard Convolution / Cost of Depthwise Separable = $1/N + 1/D_k^2$

- Significance: Depthwise separable convolutions are significantly more computationally efficient.
 - Example: If $N=100$ filters and $D_k=512$ (kernel dimension), $R \approx 0.01$. This means it performs about **100 times fewer multiplications** than standard convolution.
 - Benefit: Enables faster models, crucial for mobile and edge devices, often with minimal impact on accuracy.

Type of Convolution	Complexity
Standard	$N \times D_p^2 \times D_g^2 \times M$
Depth wise separable	$M \times D_p^2 \times (D_k^2 + N)$

MobileNet V3 Key Features

MobileNetV3 Architecture and Features



$$\text{h-swish}(x) = x * \text{ReLU6}(x + 3)/6$$

$$\text{h-sigmoid}(x) = \text{ReLU6}(x + 3)/6$$

MobileNetV3 Architecture

Input	Operator	exp size	#out	SE	NL	s
$224^2 \times 3$	conv2d	-	16	-	HS	2
$112^2 \times 16$	bneck, 3x3	16	16	-	RE	1
$112^2 \times 16$	bneck, 3x3	64	24	-	RE	2
$56^2 \times 24$	bneck, 3x3	72	24	-	RE	1
$56^2 \times 24$	bneck, 5x5	72	40	✓	RE	2
$28^2 \times 40$	bneck, 5x5	120	40	✓	RE	1
$28^2 \times 40$	bneck, 5x5	120	40	✓	RE	1
$28^2 \times 40$	bneck, 3x3	240	80	-	HS	2
$14^2 \times 80$	bneck, 3x3	200	80	-	HS	1
$14^2 \times 80$	bneck, 3x3	184	80	-	HS	1
$14^2 \times 80$	bneck, 3x3	184	80	-	HS	1
$14^2 \times 80$	bneck, 3x3	480	112	✓	HS	1
$14^2 \times 112$	bneck, 3x3	672	112	✓	HS	1
$14^2 \times 112$	bneck, 5x5	672	160	✓	HS	2
$7^2 \times 160$	bneck, 5x5	960	160	✓	HS	1
$7^2 \times 160$	bneck, 5x5	960	160	✓	HS	1
$7^2 \times 160$	conv2d, 1x1	-	960	-	HS	1
$7^2 \times 960$	pool, 7x7	-	-	-	-	1
$1^2 \times 960$	conv2d 1x1, NBN	-	1280	-	HS	1
$1^2 \times 1280$	conv2d 1x1, NBN	-	k	-	-	1

Table 1. Specification for MobileNetV3-Large. SE denotes whether there is a Squeeze-And-Excite in that block. NL denotes the type of nonlinearity used. Here, HS denotes h-swish and RE denotes ReLU. NBN denotes no batch normalization. *s* denotes stride.

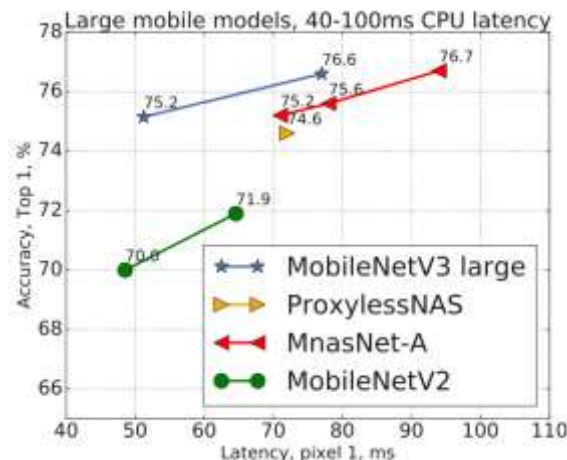
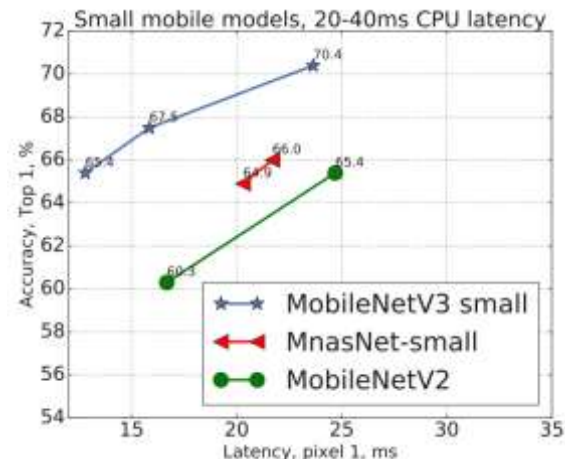
Input	Operator	exp size	#out	SE	NL	s
$224^2 \times 3$	conv2d, 3x3	-	16	-	HS	2
$112^2 \times 16$	bneck, 3x3	16	16	✓	RE	2
$56^2 \times 16$	bneck, 3x3	72	24	-	RE	2
$28^2 \times 24$	bneck, 3x3	88	24	-	RE	1
$28^2 \times 24$	bneck, 5x5	96	40	✓	HS	2
$14^2 \times 40$	bneck, 5x5	240	40	✓	HS	1
$14^2 \times 40$	bneck, 5x5	240	40	✓	HS	1
$14^2 \times 40$	bneck, 5x5	120	48	✓	HS	1
$14^2 \times 48$	bneck, 5x5	144	48	✓	HS	1
$14^2 \times 48$	bneck, 5x5	288	96	✓	HS	2
$7^2 \times 96$	bneck, 5x5	576	96	✓	HS	1
$7^2 \times 96$	bneck, 5x5	576	96	✓	HS	1
$7^2 \times 96$	conv2d, 1x1	-	576	✓	HS	1
$7^2 \times 576$	pool, 7x7	-	-	-	-	1
$1^2 \times 576$	conv2d 1x1, NBN	-	1024	-	HS	1
$1^2 \times 1024$	conv2d 1x1, NBN	-	k	-	-	1

Table 2. Specification for MobileNetV3-Small. See table 1 for notation.

MobileNet V3 Results (ImageNet Dataset)

Network	Top-1	MAdds	Params	P-1	P-2	P-3
V3-Large 1.0	75.2	219	5.4M	51	61	44
V3-Large 0.75	73.3	155	4.0M	39	46	40
MnasNet-A1	75.2	315	3.9M	71	86	61
Proxyless[5]	74.6	320	4.0M	72	84	60
V2 1.0	72.0	300	3.4M	64	76	56
V3-Small 1.0	67.4	56	2.5M	15.8	19.4	14.4
V3-Small 0.75	65.4	44	2.0M	12.8	15.6	11.7
Mnas-small [43]	64.9	65.1	1.9M	20.3	24.2	17.2
V2 0.35	60.8	59.2	1.6M	16.6	19.6	13.9

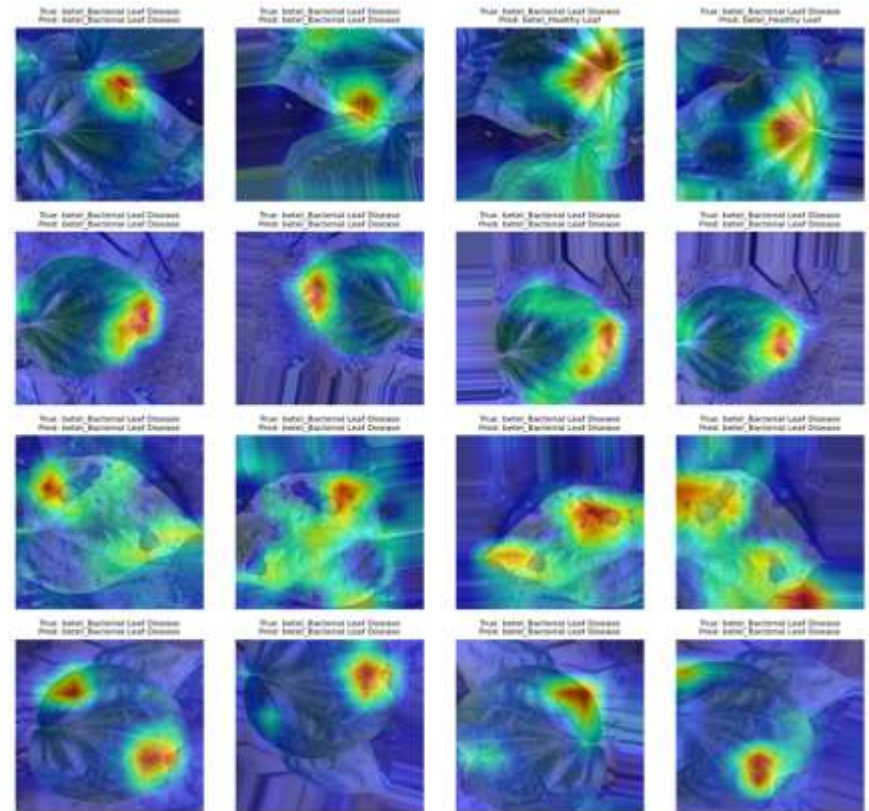
Table 3. Floating point performance on the Pixel family of phones



Attention Mechanisms in Computer Vision

Attention mechanisms have revolutionized the field of computer vision, enhancing the capability of neural networks to focus on the most relevant parts of an image. By dynamically adjusting the focus, these mechanisms mimic human visual attention, enabling more precise and efficient processing of visual information.

Attention Mechanisms selectively emphasize important features of an input image while downplaying less relevant ones. This dynamic weighting is crucial for tasks where specific parts of an image carry more significance than others, such as in object detection, image segmentation, and image captioning



Squeeze-and-Excitation (SE) Attention Mechanism

Squeeze

Global average pooling across each $H \times W$ feature map produces a $1 \times 1 \times C$ channel descriptor, embedding global context into a compact vector.

Excitation + Gating (Hard-Sigmoid)

A two-layer bottleneck ($FC \downarrow$) $\rightarrow$ ReLU $\rightarrow$ ($FC \uparrow$) transforms the C-dim descriptor.

Instead of the standard sigmoid, a serves as the gating activation for efficiency on mobile hardware.

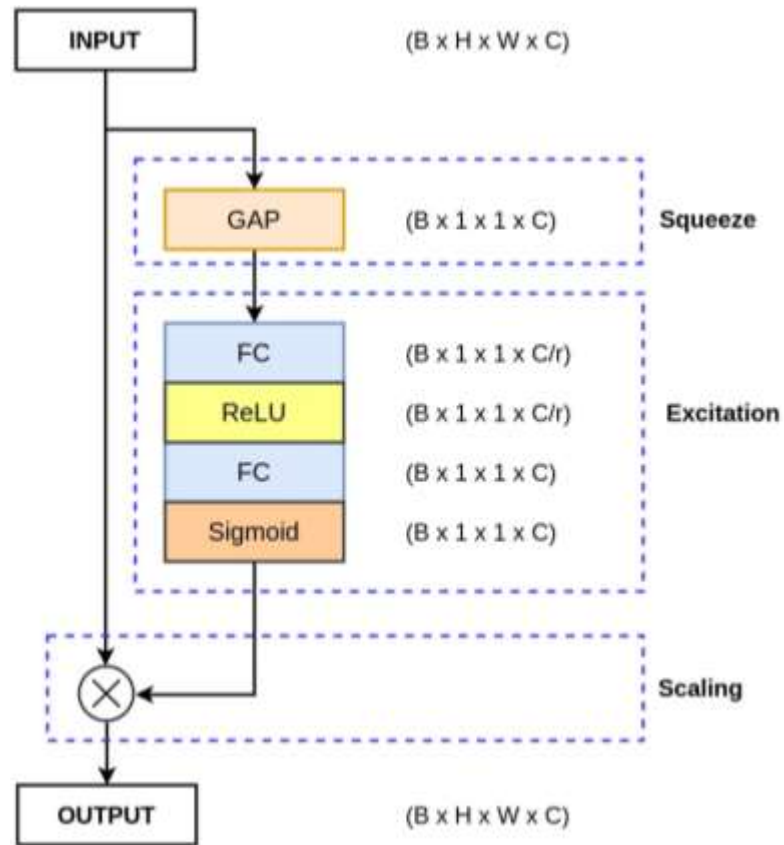
$$(\text{h-sigmoid}(x) = \text{ReLU6}(x + 3)/6)$$

Scaling

The resulting C-dim weight vector is reshaped to $1 \times 1 \times C$ and multiplied channel-wise with the original feature map, emphasizing informative channels and suppressing others.

Placement in MobileNetV3

SE blocks with hard-sigmoid gating are inserted after the depthwise convolution within each inverted residual bottleneck, balancing accuracy and latency on resource-constrained devices.

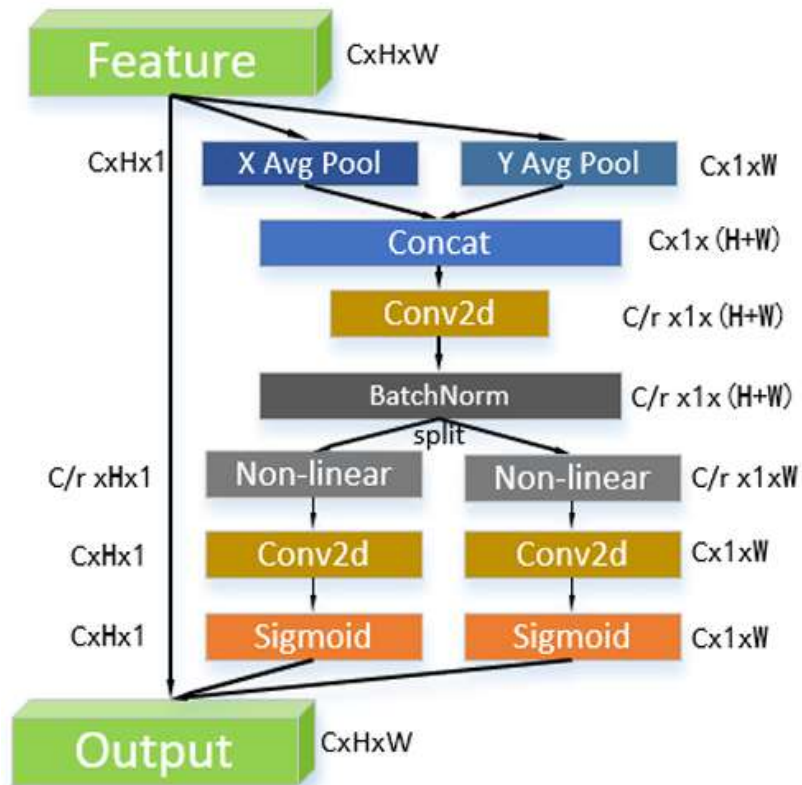


Coordinate Attention (CA)

Coordinate Attention (CA) enhances feature learning by embedding spatial positional information into channel attention. It decomposes 2D global pooling into horizontal and vertical 1D pooling to capture long-range dependencies, enabling precise localization of disease spots (e.g., small lesions on leaves). However, CA relies on Batch Normalization (BN) and channel reduction, which can degrade performance in small models and cluttered environments like mixed tea/betel crops.

Advantages:

- Spatial-Channel Fusion: Captures both where (spatial) and what (channel) to focus on.
- Long-Range Dependencies: Detects subtle disease patterns across large areas (e.g., tea leaves).
- Improved Accuracy: Boosts MobileNetV2 performance by ~2% on ImageNet.

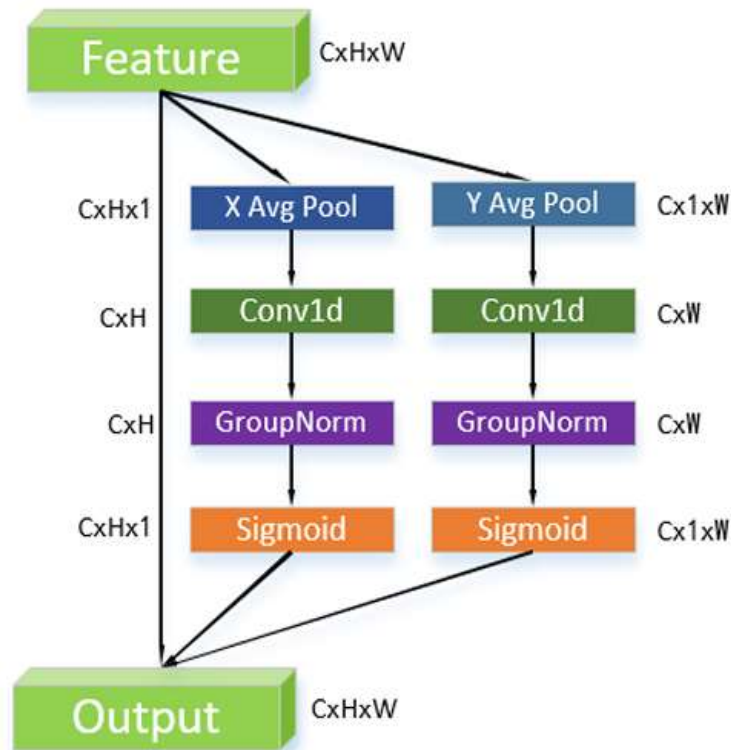


Efficient Local Attention (ELA)

Efficient Local Attention (ELA) addresses CA's limitations by replacing BN with Group Normalization (GN) and using 1D convolutions to process spatial sequences without channel reduction. This preserves direct channel relationships, improves generalization on small datasets, and enhances focus on tiny lesions in complex backgrounds. ELA's lightweight design (e.g., <0.2M parameters) ensures compatibility with mobile devices, making it ideal for Assam's small farmers growing mixed crops.

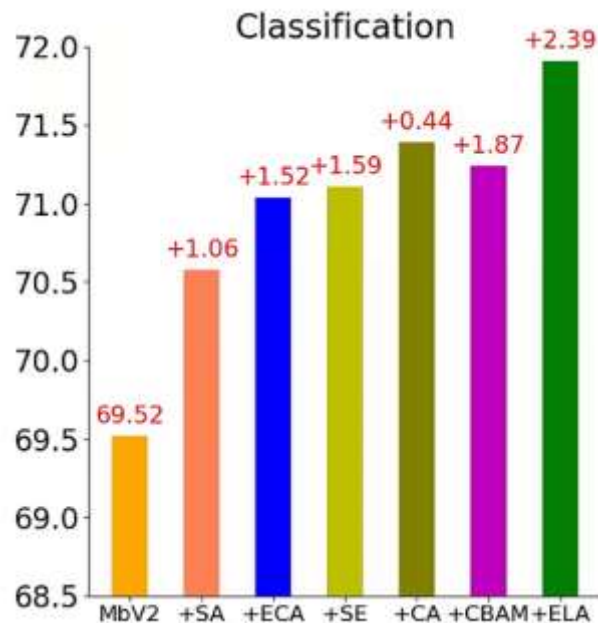
Advantages:

- No Channel Reduction: Preserves direct channel relationships for better feature learning.
- GN over BN: Robust to small batch sizes, ideal for mobile/edge devices.
- 1D Convolutions: Lightweight processing of spatial sequences (5–7x faster than 2D).
- Scalability: Four versions (ELA-T, ELA-B, ELA-S, ELA-L) for diverse architectures.



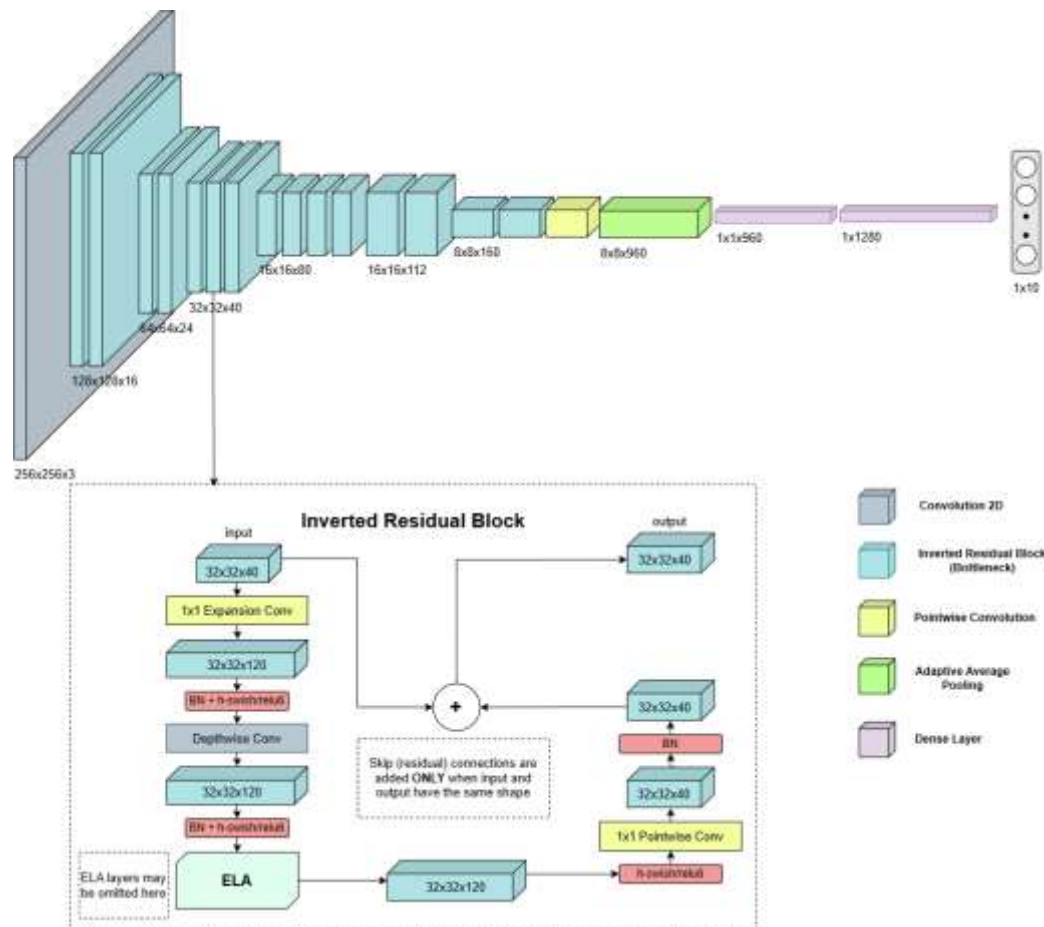
Performance Analysis of ELA

Attention Methods	Bankbones	Top-1(%)	Top-5(%)	Params(M)
MobileNetV2-1.0	MobileNetV2-1.0	69.52	89.21	3.50
+CA(r=32)	MobileNetV2-1.0	71.39	90.57	3.95
+CA(r=24)	MobileNetV2-1.0	71.56	90.54	4.09
+CA(r=16)	MobileNetV2-1.0	71.90	90.79	4.37
+ECA	MobileNetV2-1.0	71.04	90.19	3.51
+SE	MobileNetV2-1.0	71.11	90.14	3.79
+SA	MobileNetV2-1.0	70.58	90.09	3.51
+CBAM	MobileNetV2-1.0	71.24	90.42	3.79
+ELA-S(Ours)	MobileNetV2-1.0	71.52	90.57	3.79
+ELA-L(Ours)	MobileNetV2-1.0	71.91	90.68	3.90



Proposed Model

MobileNet V3 + ELA Module



Results and Analysis

Training Strategy: Two-Phase Approach

A two-phase transfer Learning strategy was applied to train the model.

- **Key Modules Always Trainable:** The Attention Mechanism Module Coordinate Attention (CA) and the Efficient Localization Attention (ELA) were trainable throughout both Phase 1 and Phase 2.
- **Phase 1:** Classifier Training: Train only the classifier layer(s) with the base model frozen.
- **Phase 2:** Fine-tuning: Unfreeze and train the last 20 layers of the base model along with the classifier.

Two-Phase Transfer Learning Strategy



Results and Analysis

Phase 1: Training the Classifier (Frozen Base)

Configuration:

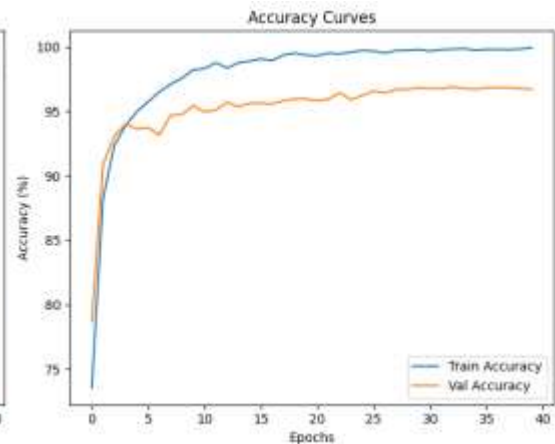
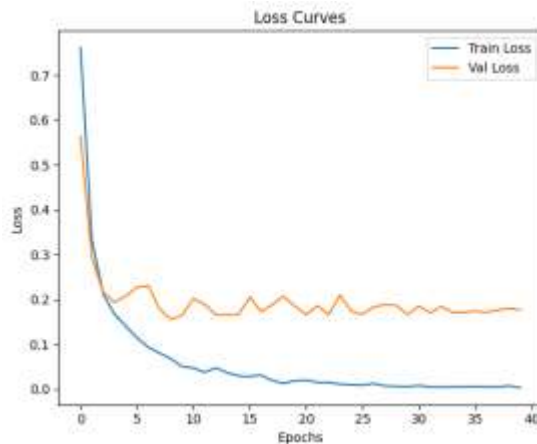
- Trainable: Classifier Layer(s) Only
- Optimizer: Adam (LR: $3e-4$)
- Scheduler: Cosine Annealing LR
- Epochs: 40
- Batch Size: 16

Results:

This phase trained the classifier to use the pre-learned features.

Best Validation Accuracy during training: **96.91%**

Final Test Accuracy after Phase 1: **97.28%**



Results and Analysis

Phase 2: Fine-tuning (Unfreezing Layers)

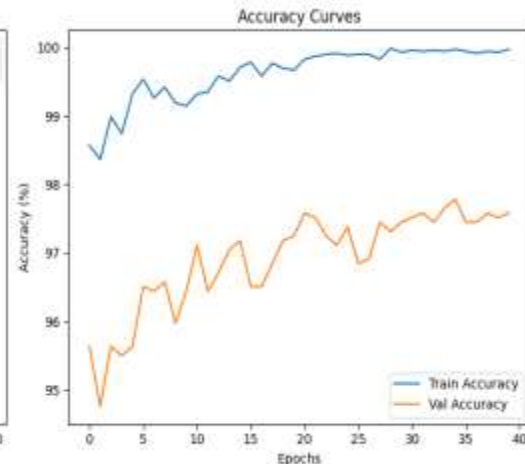
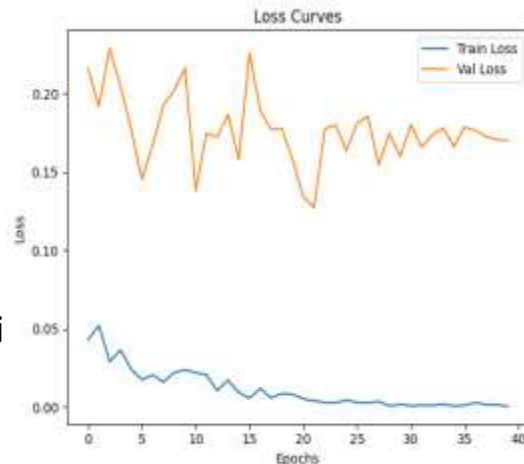
Configuration:

- Trainable: Last 20 Base Layers + Classifier
- Optimizer: Adam (LR: $3e-4$)
- Scheduler: Cosine Annealing LR
- Epochs: Additional 40 epochs for fine-tuning
- Batch Size: 16

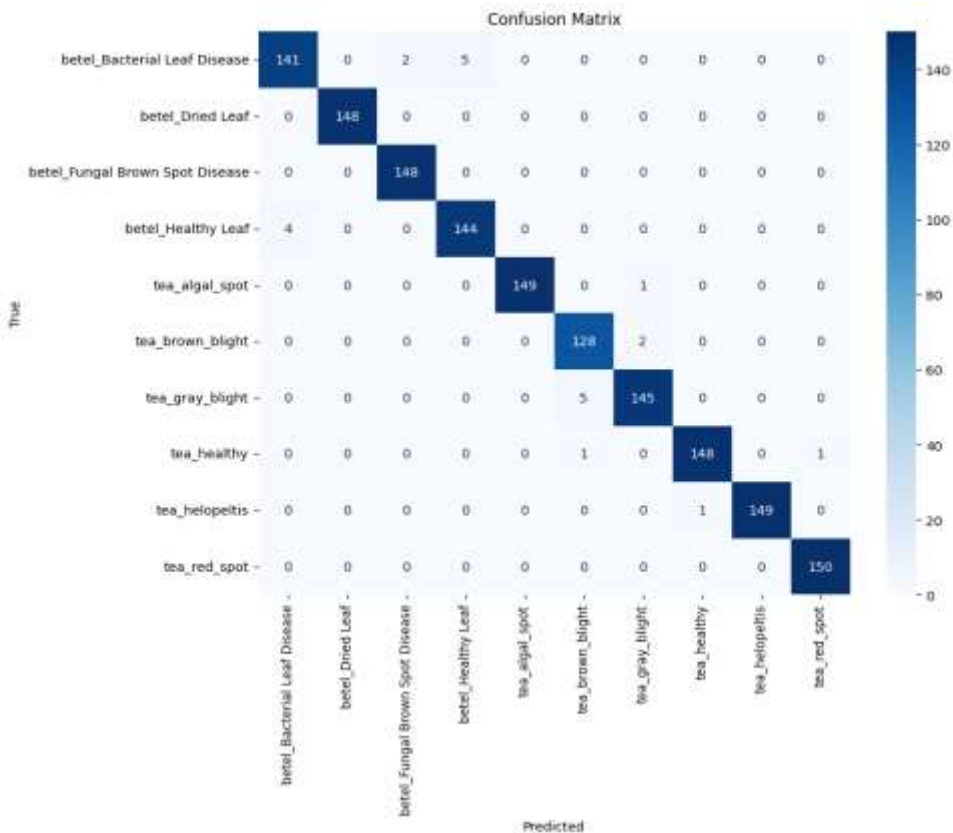
Results:

Fine-tuning the later layers further improved performance.

- Best Validation Accuracy during training: **97.78%**
- Final Test Accuracy after Phase 2: **98.51%**



Results and Analysis



Test Accuracy: 98.51%

Classification Report:

	precision	recall	f1-score	support
betel_Bacterial Leaf Disease	0.97	0.95	0.96	148
betel_Dried Leaf	1.00	1.00	1.00	148
betel_Fungal Brown Spot Disease	0.99	1.00	0.99	148
betel_Healthy Leaf	0.97	0.97	0.97	148
tea_algal_spot	1.00	0.99	1.00	150
tea_brown_blight	0.96	0.98	0.97	130
tea_gray_blight	0.98	0.97	0.97	150
tea_healthy	0.99	0.99	0.99	150
tea_helopeltis	1.00	0.99	1.00	150
tea_red_spot	0.99	1.00	1.00	150
accuracy			0.99	1472
macro avg	0.98	0.99	0.98	1472
weighted avg	0.99	0.99	0.99	1472

Results and Analysis

True: tea_gray_blight
Pred: tea_gray_blight



True: tea_brown_blight
Pred: tea_brown_blight



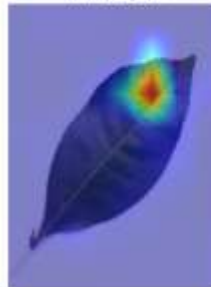
True: betel_Fungal Brown Spot Disease
Pred: betel_Fungal Brown Spot Disease



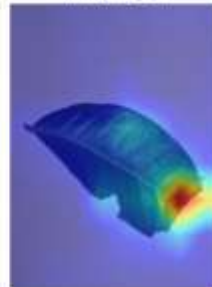
True: tea_red_spot
Pred: tea_red_spot



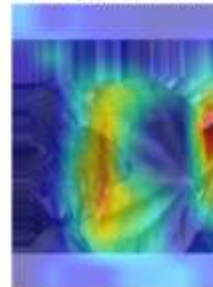
True: tea_gray_blight
Pred: tea_gray_blight



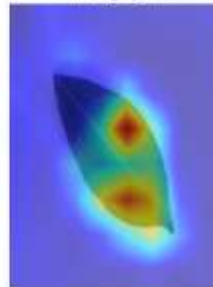
True: tea_brown_blight
Pred: tea_brown_blight



True: betel_Fungal Brown Spot Disease
Pred: betel_Fungal Brown Spot Disease



True: tea_red_spot
Pred: tea_red_spot



True: betel_Bacterial Leaf Disease
Pred: betel_Bacterial Leaf Disease



True: betel_Bacterial Leaf Disease
Pred: betel_Bacterial Leaf Disease



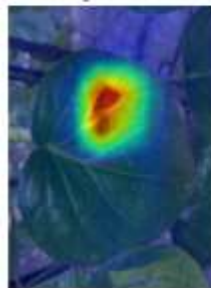
True: tea_gray_blight
Pred: tea_gray_blight



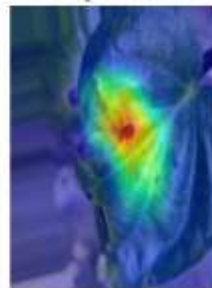
True: betel_Bacterial Leaf Disease
Pred: betel_Bacterial Leaf Disease



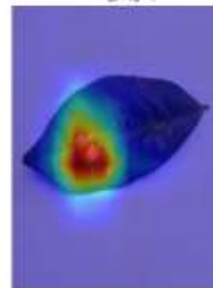
True: betel_Bacterial Leaf Disease
Pred: betel_Bacterial Leaf Disease



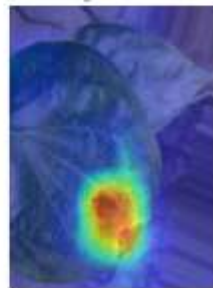
True: betel_Bacterial Leaf Disease
Pred: betel_Bacterial Leaf Disease



True: tea_gray_blight
Pred: tea_gray_blight



True: betel_Bacterial Leaf Disease
Pred: betel_Bacterial Leaf Disease



Results and Analysis

Models	Params (M)	Test Accuracy (%)
Simple CNN	10.3	88.25
EfficientNet B0	5.3	97.36
Mnasnet-A1	3.11	93.00
MobileNet V3 Baseline	4.21	97.49
MobileNet V3 + CA	2.99	97.62
MobileNet V3 + ELA (our)	2.87	98.51

Conclusion

- A lightweight and efficient model using **MobileNetV3 integrated with Efficient Local Attention (ELA)** was proposed for accurate classification of tea and betel leaf diseases.
- The proposed model was evaluated against other architectures like baseline MobileNetV3, MobileNetV3 with fine-tuning, MnasNet, EfficientNet-B0, and a custom CNN.
- MobileNetV3 + ELA consistently delivered the highest test accuracy** while maintaining a low parameter count, making it optimal for real-world deployment.
- The model's **compact size and high performance** make it suitable for **mobile or edge devices**, helping small-scale farmers detect leaf diseases without complex equipment.
- This approach can significantly contribute to **early disease management** and improved agricultural productivity in resource-constrained settings.

Future Improvements

Mobile Application Development:

Plan to deploy the trained model in Android and iOS mobile apps to enable easy, on-the-go disease detection for farmers.

Integration with IoT and Drones:

Incorporate the model into IoT devices or agricultural drones for real-time monitoring and detection of leaf diseases in the field.

Expanding Dataset:

Enhance the model by including more diverse and newly emerging disease categories in tea and betel leaves for broader applicability and robustness.

REFERENCE

- **Mohanty, S. P., Hughes, D. P., & Salathé, M.** (2016). Using deep learning for image-based plant disease detection
- **Sladojevic, S., Arsenovic, M., Anderla, A., Culibrk, D., & Stefanovic, D.** (2016). Deep neural networks based recognition of plant diseases by leaf image classification.
- **Too, E. C., Yujian, L., Njuki, S., & Yingchun, L.** (2019). A comparative study of fine-tuning deep learning models for plant disease identification.
- **Howard, A., Sandler, M., et al.** (2019). Searching for MobileNetV3.
- **Hou, Q., Zhou, D., Feng, J.** (2021). Coordinate Attention for Efficient Mobile Network Design.
- **Wang, Q., Wu, B., Zhu, P., Li, P., Zuo, W., & Hu, Q.** (2020). ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks.
- **Amara, J., Bouaziz, B., & Algergawy, A.** (2017). A Deep Learning-based Approach for Banana Leaf Diseases Classification.
- **Zhang, Y., Chen, M., Liu, Y., & Zhang, Y.** (2022). Tomato leaf disease identification using attention-based EfficientNet.

THANK YOU